THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 09

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** Student A is correct	$\sqrt{4\times3}$
2	**(b)** Incorrect; $\sqrt{9}=3$ is rational	$\sqrt{9}=3$
3	**(b)** x^2-3x-4	Sum 3, product -4
4	**(a)** downwards	$a<0$
5	**(b)** $(3,1)$	Adding: $3x=9$
6	**(c)** $12, 8$	$x+y=20,\ x-y=4 \Rightarrow 12, 8$
7	**(b)** ±3	$x^2=9$
8	**(d)** $-\frac{9}{4}$	$\frac{3}{4}-k-3=0$
9	**(d)** −4	$d=-3$
10	**(a)** 4	Coefficient of n
11	**(b)** 27 m	$\frac{6}{4}=\frac{h}{18}$
12	**(b)** Incorrect — the included angles must also be equal (SAS)	SAS needs the included angle
13	**(a)** 0	Same terms
14	**(b)** $6\sqrt{2}$ cm	OAPB is a square; OP is its diagonal
15	**(b)** Incorrect — it is πr	Half of $2\pi r$
16	**(c)** 60°	$\frac{\theta}{360}\times36\pi=6\pi$
17	**(d)** 440 cm²	$2\times\frac{22}{7}\times7\times10$
18	**(b)** 15	$\frac{10+20}{2}$
19	**(b)** x_i-a	Deviation from assumed mean
20	**(a)** $\frac{2}{11}$	2 B's in 11 letters

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$=5(7\times6\times4\times3\times2\times1+1)=5\times1009$; it has a factor 5 other than 1 and itself ⇒ composite.	2
22	$(2s-1)^2$ ⇒ zeroes $\frac{1}{2},\frac{1}{2}$. Sum $1=\frac{4}{4}$ ✓; product $\frac{1}{4}$ ✓	1+1
23	$D=48-48=0$ ⇒ equal real roots; $x=\frac{4\sqrt{3}}{6}=\frac{2}{\sqrt{3}}$ (both)	1+1
24	(i) Any two squares; any two equilateral triangles (ii) A square and a rectangle; a square and a rhombus (non-square)	1+1
25	$(x-2)^2+25=(x+2)^2+81 \Rightarrow -8x=56 \Rightarrow x=-7$ ⇒ $(-7,0)$	1+1
26	$\frac{\frac{1}{2}-\frac{2}{\sqrt{3}}}{\frac{1}{2}+\frac{2}{\sqrt{3}}}=\frac{3\sqrt{3}-4}{3\sqrt{3}+4}=\frac{43-24\sqrt{3}}{11}$	1+1
27	$\sin30^\circ=\frac{h}{20}\Rightarrow h=10$ m	1+1
28	The radius at the point of contact is ⊥ to the tangent; only one perpendicular can be drawn at a point on a line ⇒ the perpendicular coincides with the radius, so it passes through O.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,6),(6,0);(2,0),(0,-2)$. Intersect at $(4,2)$ ⇒ consistent	2+1
30	$20+\frac{9-5}{18-5-7}\times10=26.67$	1+1+1
31	From the end: $201-11\times2=179$	1+1+1
32	$\triangle OPT\cong\triangle OQT$ (RHS: $OP=OQ$, OT common, right angles) ⇒ $\angle OTP=\angle OTQ$	1+1+1
33	Sector ≈209.33; equilateral triangle $=\frac{\sqrt{3}}{4}\times400=173$ ⇒ segment ≈36.33 cm²	1+1+1
34	48 cards: $\frac{4}{48}=\frac{1}{12}$; $\frac{24}{48}=\frac{1}{2}$; 0	1+1+1

35	$d = \frac{50}{\tan 30^\circ} = 50\sqrt{3}$; hill $= d\tan 60^\circ = 150$ m	1+1+1
	OR: (i) $40\tan 30^\circ = \frac{40}{\sqrt{3}} \approx 23.09$ m (ii) top of tank at 40 m ⇒ depth $= 40 - 23.09 \approx 16.91$ m	1½+1½
36	$\frac{980+60p}{20+p} = 50 \Rightarrow 10p = 20 \Rightarrow p = 2$	1+1+1
	OR: Frequencies 5, 7, 13, 13, 7, 5; $\frac{n}{2} = 25$ ⇒ class 20–30: $20 + \frac{25-12}{13} \times 10 = 30$	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\sqrt{64+36} = 10$ units (ii) $P = \left(\frac{10+6}{4}, \frac{9+9}{4}\right) = (4, 4.5)$ (iii) $M = (6, 6)$ (iv) $MS = MH = \sqrt{16+9} = 5$ ✓	1+2+1+1
	OR: (i) $D = \left(\frac{5}{2}, \frac{11}{2}\right)$, $E = \left(\frac{11}{2}, 4\right)$ (ii) $DE = \sqrt{9+\frac{9}{4}} = \frac{3\sqrt{5}}{2}$, $BC = \sqrt{36+9} = 3\sqrt{5}$ ✓ (iii) Mid-point theorem	2+2+1
38	(i) 10 cm (ii) $\sin P = \frac{QR}{PR} = \frac{3}{5}$, $\cos R = \frac{QR}{PR} = \frac{3}{5}$ (iii) Both equal $\frac{QR}{PR}$; P and R are complementary (iv) $\frac{6}{8} \times \frac{8}{6} = 1$	1+2+1+1
	OR: LHS $= \frac{(\sec\theta - 1) + (\sec\theta + 1)}{\sqrt{\sec^2\theta - 1}} = \frac{2\sec\theta}{\tan\theta} = \frac{2}{\cos\theta} \cdot \frac{\cos\theta}{\sin\theta} = 2\csc\theta$	2+2+1
39	(i) 25 cm (ii) $550 + 308 = 858$ cm² (iii) ₹429 (iv) $1232 + 718.67 = 1950.67$ cm³	1+2+1+1
	OR: (i) 5.775 cm³ (ii) 0.154 cm³ (iii) 5.929 cm³ (iv) $l = 1.25$ cm; $\pi rl = 1.375$ cm²	1+1+1+2
40	(i) 800–1000 (ii) $800 + \frac{15-10}{30-10-12} \times 200 = 925$ (iii) $a = 900$, $\Sigma fd = -2800 - 2000 + 0 + 2400 + 2400 = 0$ ⇒ mean = ₹900	1+2+2
	OR: (i) cf 5, 15, 33 ⇒ $20 + \frac{25-15}{18} \times 10 = 25.56$ (ii) $a = 25$, $\Sigma fu = -10 - 10 + 0 + 12 + 10 = 2$ ⇒ 25.4 (iii) He probably took the class 30–40 (after the median class) or used n instead of $\frac{n}{2}$	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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